Orbital Stability of the Hydrogen 1s Electron under Tau-Neutrino Re-energisation
Updated Reaction Parameters: $E_{\tau_\nu} = 1.975$ eV, $E_{\nu_\mu} = 1.795$ eV, $\Delta E_e = 0.18$ eV

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Abstract

In the semiclassical picture of the hydrogen atom, an electron on a circular Bohr orbit radiates via the Larmor mechanism and should collapse onto the nucleus within $T_0 \approx 1.56 \times 10^{-11}$ s. Quantum mechanics removes the paradox by postulating non-radiating stationary states. The TauTimp (Onduscular) framework instead proposes a dynamical equilibrium in which the electron is continuously replenished by discrete energy quanta delivered through tau-neutrino scattering, $\tau_\nu + e^- \to e^- + \nu_\mu + \Delta E_e$. Using the updated total energies $E_{\tau_\nu} = 1.975$ eV and $E_{\nu_\mu} = 1.795$ eV, the energy transferred to the electron per collision is $\Delta E_e = 0.180$ eV. Equating the time-averaged injected power to the Larmor loss at the Bohr radius yields a required quantum arrival rate of $\dot{N} \approx 1.62 \times 10^{12}$ s^{-1} per hydrogen electron, i.e. one quantum every ≈ 0.62 ps, or about 25 quanta per classical collapse period.

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1. Introduction and Scope

The objective of this note is narrow and quantitative: **given the updated tau-neutrino reaction parameters, how many energy quanta per second must a ground-state ($1s$) hydrogen electron absorb in order to remain on its orbit?**

The calculation proceeds in three steps:

1. Compute the classical (Larmor) radiative power lost by an electron on a circular orbit at the Bohr radius a_0.
2. Fix the energy delivered to the electron per tau-neutrino collision from conservation of energy in the updated reaction.
3. Divide the loss rate by the energy per quantum to obtain the required injection rate $\dot{N}$, and express it as an inter-collision interval.

For simplicity the electron is treated as a point charge on a circular Keplerian orbit at fixed $r = a_0$ (the "$1s$" orbital in the correspondence limit), with no screening, no recoil of the proton, and time-averaged energy balance.

2. Physical Constants and Orbital Parameters

Quantity	Symbol	Value
Electron mass	m_e	9.109×10^{-31} kg
Elementary charge	e	1.602×10^{-19} C
Vacuum permittivity	ε_0	8.854×10^{-12} F/m
Speed of light	c	2.998×10^{8} m/s
Bohr radius	a_0	5.292×10^{-11} m

Quantity	Symbol	Value
Energy unit conversion	1 eV	1.602×10^{-19} J

Circular-orbit force balance (centripetal = Coulomb):

$$m_e \frac{V^2}{r} = \frac{e^2}{4\pi\varepsilon_0 r^2} \quad \Rightarrow \quad V^2 = \frac{e^2}{4\pi\varepsilon_0 m_e r}.$$

At $r = a_0$: $V = 2.188 \times 10^6$ m/s ($\beta_e = V/c \approx 1/137$). The total mechanical energy follows from the virial theorem:

$$E_{\text{orb}}(r) = -\frac{e^2}{8\pi\varepsilon_0 r}, \qquad E_{\text{orb}}(a_0) = -13.6 \text{ eV}.$$

3. Classical Energy-Loss Rate (Larmor Mechanism)

3.1 Radiated power

A non-relativistic accelerated charge radiates at the Larmor rate. For a circular orbit the centripetal acceleration is $a = V^2/r$, hence

$$P_{\text{Larmor}} = \frac{e^2 a^2}{6\pi\varepsilon_0 c^3} = \frac{e^6}{96\pi^3 \varepsilon_0^3 c^3 m_e^2 r^4}.$$

Evaluated at $r = a_0$:

$$\boxed{P_{\text{Larmor}}(a_0) = 4.666 \times 10^{-8} \text{ W} = 2.912 \times 10^{11} \text{ eV/s}.}$$

This is the energy the electron must receive, on time average, to hold its orbit.

3.2 Classical collapse time

Integrating the radiative spiral from $r = a_0$ to $r = 0$:

$$T_0 = \frac{4\pi^2 \varepsilon_0^2 c^3 m_e^2 a_0^3}{e^4} \approx 1.557 \times 10^{-11} \text{ s}.$$

4. Updated Tau-Neutrino Reaction

4.1 Reaction and energy budget

$$\tau_\nu + e^- \rightarrow e^- + \nu_\mu + \Delta E_e.$$

Quantity	Symbol	Value
Tau-neutrino total energy (incident)	E_{τ_ν}	1.975 eV
Muon-neutrino total energy (product)	E_{ν_μ}	1.795 eV
Energy deposited into the electron	ΔE_e	0.180 eV

Conservation of energy (electron kinetic recoil absorbed into ΔE_e):

$$\Delta E_e = E_{\tau_\nu} - E_{\nu_\mu} = 1.975 - 1.795 = 0.180 \text{ eV}.$$

Each collision is therefore a **quantum of** 0.180 eV transferred to the bound electron.

4.2 Kinematics of the incident tau neutrino

With rest energy $m_{\tau_\nu} c^2 = 0.188$ eV and $E_{\tau_\nu} = 1.975$ eV:

$$\gamma = \frac{E_{\tau_\nu}}{m_{\tau_\nu} c^2} = \frac{1.975}{0.188} = 10.505, \qquad \beta = \sqrt{1 - \gamma^{-2}} = 0.99546 \quad (v = 99.55\%\, c).$$

$$pc = \sqrt{E_{\tau_\nu}^2 - (m_{\tau_\nu} c^2)^2} = 1.966 \text{ eV}, \qquad \lambda_{\tau_\nu} = \frac{hc}{pc} = \frac{1239.8 \text{ eV} \cdot \text{nm}}{1.966 \text{ eV}} \approx 630.6 \text{ nm}.$$

4.3 Kinematics of the product muon neutrino

In the ultra-relativistic limit ($m_{\nu_\mu} \approx 0$):

$$p_{\nu_\mu} c \approx E_{\nu_\mu} = 1.795 \text{ eV}, \qquad \lambda_{\nu_\mu} = \frac{1239.8}{1.795} \text{ nm} \approx 690.7 \text{ nm}.$$

Both the incident and the emitted neutrino lie in the visible red band (630.6 nm and 690.7 nm respectively).

5. Required Quantum Arrival Rate for 1s Stability

5.1 Steady-state condition

Orbital stability requires the time-averaged injected power to equal the Larmor loss:

$$\dot{N}\, \Delta E_e = |P_{\text{Larmor}}(a_0)|,$$

where $\dot{N}$ is the number of 0.180 eV quanta absorbed per second. Solving:

$$\boxed{\dot{N} = \frac{|P_{\text{Larmor}}(a_0)|}{\Delta E_e} = \frac{2.912 \times 10^{11} \text{ eV/s}}{0.180 \text{ eV}} \approx 1.618 \times 10^{12} \text{ s}^{-1}.}$$

5.2 Inter-collision interval

$$\Delta t_{\text{inject}} = \frac{1}{\dot{N}} \approx 6.18 \times 10^{-13} \text{ s} \approx 0.618 \text{ ps}.$$

5.3 Quanta per classical collapse period

$$\frac{T_0}{\Delta t_{\text{inject}}} = T_0\, \dot{N} \approx 25.2.$$

Within one classical collapse time the electron must intercept about **25 quanta** of 0.180 eV each; their cumulative energy, $25 \times 0.180 \approx 4.5$ eV, is exactly the energy the orbit would have radiated away over T_0 ($|P_{\text{Larmor}}|\, T_0 \approx 4.53$ eV).

5.4 Energy-balance verification

Quantity	Value
Larmor power loss, $\lvert P_{\text{Larmor}}(a_0)\rvert$	2.912×10^{11} eV/s
Required arrival rate, $\dot{N}$	1.618×10^{12} s^{-1}
Energy per quantum, ΔE_e	0.180 eV
Injected power, $\dot{N}\, \Delta E_e$	2.912×10^{11} eV/s
Balance ratio	1.000

6. Results Summary

Parameter	Symbol	Value
Tau-neutrino energy (incident)	E_{τ_ν}	1.975 eV
Muon-neutrino energy (product)	E_{ν_μ}	1.795 eV
Energy per collision to electron	ΔE_e	0.180 eV
Larmor power at a_0	$\lvert P_L\rvert$	2.912×10^{11} eV/s (4.67×10^{-8} W)
Required quanta per second	$\dot{N}$	$\mathbf{1.618 \times 10^{12}\ s^{-1}}$
Inter-collision interval	Δt	6.18×10^{-13} s (0.618 ps)
Classical collapse time	T_0	1.557×10^{-11} s
Quanta per collapse period	$T_0/\Delta t$	≈ 25
Tau-neutrino speed	β	$0.9955\, c$
Tau-neutrino wavelength	λ_{τ_ν}	630.6 nm (red)
Muon-neutrino wavelength	λ_{ν_μ}	690.7 nm (deep red)

7. Discussion

Answer to the guiding question. To keep the hydrogen electron on the $1s$ orbital against Larmor radiation, the TauTimp mechanism must deliver

$$\dot{N} \approx 1.62 \times 10^{12} \text{ quanta per second}$$

of 0.180 eV each, i.e. one absorbed tau-neutrino collision roughly every 0.62 ps.

Magnitude of the deposited quantum. $\Delta E_e = 0.18$ eV is about 7 $k_B T$ at room temperature ($k_B T \approx 0.0259$ eV), small enough to remain spectroscopically inconspicuous against thermal broadening, yet, when delivered at the rate above, exactly sufficient to offset the classical loss.

Spectral prediction. With the updated energies both neutrinos fall in the visible red (λ=631 nm incident, λ=691 nm emitted), a shift from the near-infrared assignment of earlier parameter sets and a distinguishing, falsifiable feature of this version.

Simplifying assumptions. Circular orbit at fixed a_0; non-relativistic Larmor formula; proton recoil neglected; perfect time-averaged balance; unit absorption probability per "counted" collision (the geometric/cross-section suppression that sets the true incident neutrino flux is outside the present scope). The figure $\dot{N}$ is thus the rate of *effective* energy quanta, not the raw neutrino fluence.

8. Conclusion

Under the updated reaction $\tau_\nu + e^- \rightarrow e^- + \nu_\mu + 0.18$ eV, orbital energy balance for the hydrogen $1s$ electron fixes the required quantum arrival rate at $\dot{N} \approx 1.618 \times 10^{12}$ s^{-1} (≈ 25 quanta per classical collapse period). This is the central quantitative output of the TauTimp stabilisation hypothesis at the present parameter values.

References

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